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Deficiency symptoms: Chlorosis, anthocyanin formation, stunted growth, rolling of leaf tip and reduced nodulation in legumes.

NPK Fertilizers

It consists of nitrogen, phosphate with potassium in different proportions. The number labelled on the bags as 15-15-15 indicates N, P & K in equal proportions.

Chelating Agents

EDTA (Chemical Chelating Agent)

Plants which are growing in alkaline soil when supplied with all nutrients including iron will show iron deficiency. To rectify this, we have to make iron into a soluble complex by adding a chelating agent like EDTA (Ethylene Diamine Tetra Acetic acid) to form Fe-EDTA.

Siderophores (Biological Chelating agent)

Siderophores (iron carriers) are Iron chelating agents produced by bacteria. They are used to chelate ferric Iron (Fe^{3+}) from environment and host.

12.3 Functions, mode of absorption and deficiency symptoms of micronutrients

Micronutrients even though required in trace amounts are essential for the metabolism of plants. They play key roles in many plants. Example: Boron is essential for translocation of sugars, molybdenum is involved in nitrogen metabolism and zinc is needed for biosynthesis of auxin. Here, we will study about the role of micro nutrients, their functions, their mode of absorption, deficiency symptoms and deficiency diseases.

1. **Iron (Fe):** Iron is required lesser than macronutrient and larger than micronutrients, hence, it can be placed in any one of the groups. Iron is an essential element for the synthesis of chlorophyll and carotenoids. It is the component of cytochrome, ferredoxin, flavoprotein, formation of chlorophyll, porphyrin, activation of catalase, peroxidase enzymes. It is absorbed as ferrous (Fe^{2+}) and ferric (Fe^{3+}) ions. Mostly fruit trees are sensitive to iron.

   Deficiency: Interveinal Chlorosis, formation of short and slender stalk and inhibition of chlorophyll formation.

2. **Manganese (Mn):** Activator of carboxylases, oxidases, dehydrogenases and kinases, involved in splitting of water to liberate oxygen (photolysis). It is absorbed as manganous (Mn^{2+}) ions.

   Deficiency: Interveinal chlorosis, grey spot on oats leaves and poor root system.

3. **Copper (Cu):** Constituent of plastocyanin, component of phenolases, tyrosinase, enzymes involved in redox reactions, synthesis of ascorbic acid, maintains carbohydrate and nitrogen balance, part of oxidase and cytochrome oxidase. It is absorbed as cupric (Cu^{2+}) ions.